Eigenfaces

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Introduction

- **Image recognition** is one of the most important applications of modern technology.
- Linear algebra plays a key role in this process.
- By using eigenvectors, we can analyze and compare images effectively.
- **Goal:** Given a new face, we want to find the most similar face in a dataset of famous people.

Faces and Matrices

- Each image is compressed in order to have 120x80 pixels in black & white (that the values 120 and 80 are arbitrary, a different definition can be chosen)
- Convert each image (120x80) into a row vector of length 9600:

$$\text{Image} = \begin{bmatrix} row_1 \\ row_2 \\ \vdots \\ row_{120} \end{bmatrix}$$

$$\text{Vectorized image} = \begin{bmatrix} row_1 & row_2 & \cdots & row_{120} \end{bmatrix}$$

- Vectorized images of the dataset: $B_1, B_2, \ldots, B_m$
- Form the matrix

$$B = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_m \end{bmatrix}$$

- In our case, m is much smaller than 9600, so B is a short and wide matrix

Correlation and eigenvectors

- The correlation matrix is $C = B^T B$
- Compute the significant eigenvectors of C:

$$C\, v_i = \lambda_i v_i, \quad i = 1, \ldots, m$$

where $\lambda_1, \ldots, \lambda_m$ are the greatest eigenvectors of C
- Form $V = [v_1, v_2, \ldots, v_m]$
- To project a picture F (stored as a row vector):

$$\text{proj}(F) = F\, V,$$

- To reconstruct the projected image, $F\, V\, V^T$

Figures

Fig.1 Faces in the dataset

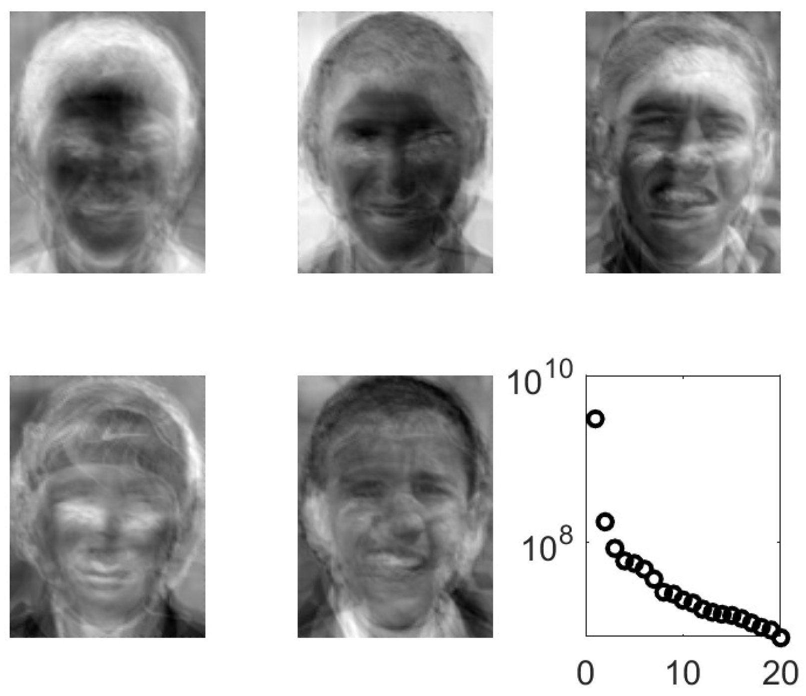

Fig.2 First 5 eigenvectors reconstructed as pictures

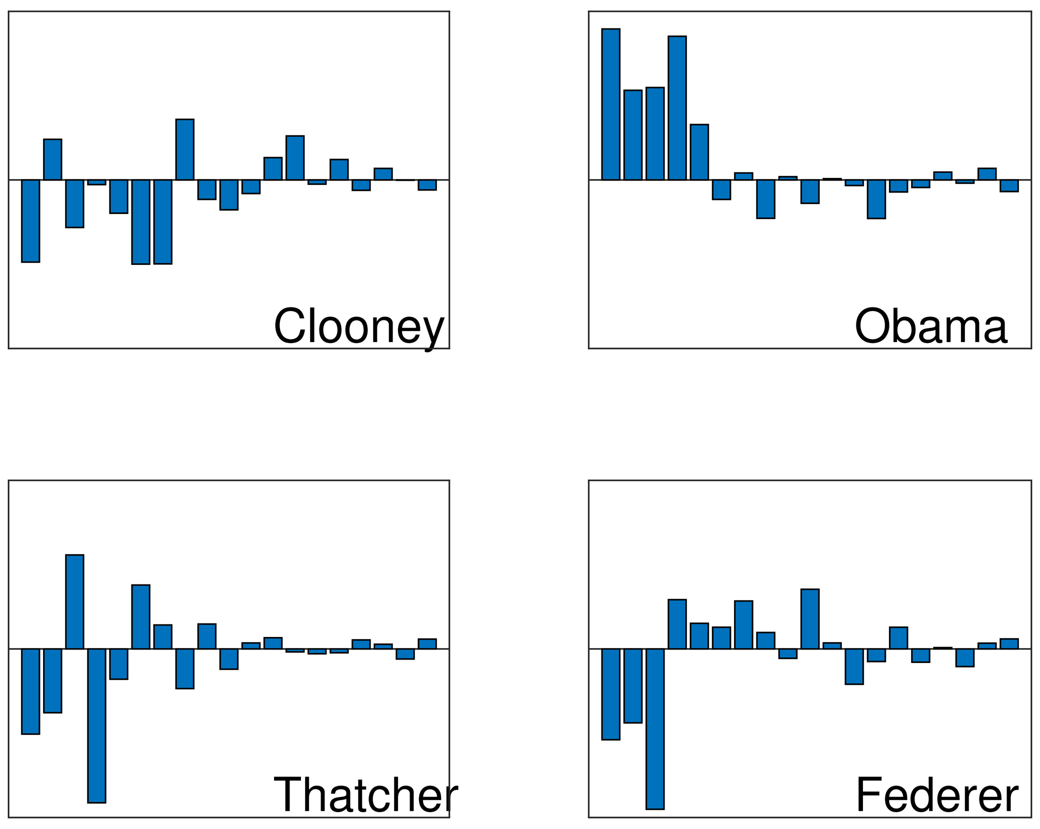

Fig.3 Components of the projection of the dataset faces

Fig. 4 Projection of a new image

Measuring Similarities

- Measure of distance between two figures F_1, F_2:

$$\|F_1\, V - F_2\, V\|_2$$

- We can use this to compare a new picture with the ones from the dataset
- Small measure means close resemblance

Results

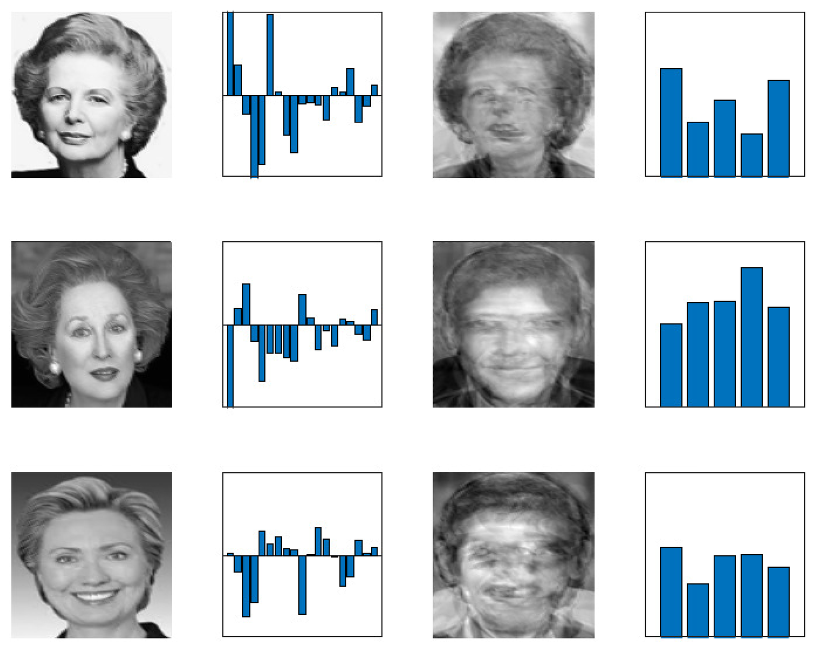

Fig.5 Meryl Streep, acting as Margaret Tatcher, is less similar to Margaret Tatcher than Hilary Clinton!

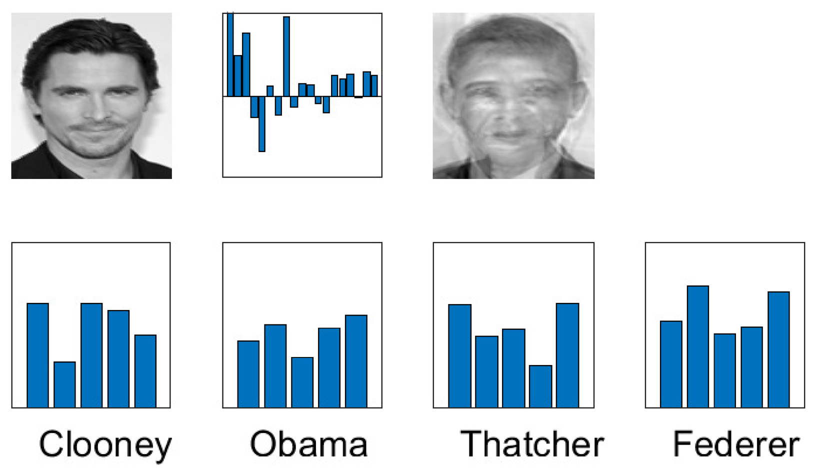

Fig.6 Comparison with a new picture (Christian Bale)

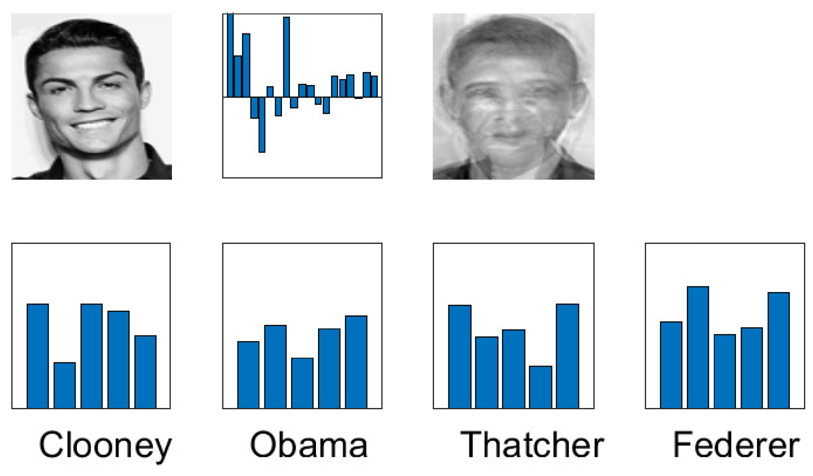

Fig.7 Comparison with a new picture (Cristiano Ronaldo)

Conclusions

- Linear algebra provides many tools for image recognition
- Spectral properties, such as eigenvalues and eigenvectors, are able to detect core features of a picture
- More sophisticated techniques yield better results (tensors, AI, Fourier transforms,...)